

Eric M. Truong

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TECHNICAL SKILLS

Programming Languages: C (7 years), C++ (4 years), Python (1 year), Shell scripting — bash and sh (5 years), Assembly for MIPS32 (3 months), JavaScript (3 months)

Operating Systems: RHEL, Unix, Linux, Ubuntu, Raspberry Pi OS, FreeRTOS, Poky

Single-board MCUs: PSoC-6, ESP32, i.MX 8M Nano

Virtualization: VMware, VirtualBox

Cloud Computing: Oracle Cloud

Web servers: Nginx

Project Management: git, Jira, Confluence, Helix

Yocto Project: Bitbake, Recipe writing (.bb + .bbappend), U-Boot

Miscellaneous: ssh, scp, vim, tmux, grep, regex, Jenkins, Yocto Project, Oscilloscopes, Multimeters

PROFESSIONAL EXPERIENCE

Cepheid

Sunnyvale, CA

Embedded Software Engineer

August 2023 – Present, Full-time

- Maintained and enhanced Yocto Linux image to work with changing proprietary hardware and firmware requirements.
- Modified U-Boot stop string code to resemble a typical password prompt to enable intuitive developer/field tech access to U-Boot command line.
- Implemented a secure Chain of Trust across U-Boot and the Linux kernel; configured cryptographic image signing and verification to prevent unauthorized firmware execution.
- Modified Device Tree source files to adjust memory mapping as part of migration to a new SOM.
- Designed and maintained Jenkins CI/CD pipelines to automate the building and archiving of C/C++ firmware, Yocto-based Linux images, and internal engineering tools.
- Engineered a fully automated flashing method for embedded Linux images on System on Modules (SOMs), eliminating manual bottlenecks and accelerating manufacturing processes by over 15x.
- Used the Yocto Project to customize embedded Linux distributions and make them work with our processors and peripherals.
- Wrote robust C/C++ firmware for the GeneXpert modules (medical diagnostic machines); Asymmetric Multi-processing: 1 core FreeRTOS, 4 cores Embedded Linux; Debugging done via GDB over JTAG and/or via log files.
- Rolled out a remote firmware update process that can handle configuration, binaries, and Linux upgrades (implemented the on-device scripts).

Western Digital Corporation

Milpitas, CA

Software Engineer

August 2022 – May 2023, Full-time

- Brought up automated testing rigs that flash firmware builds onto physical SSDs and run functional tests (Pytest) without human intervention.
- Developed, ported, and maintained Python-based Continuous Integration scripts.
- Brought up Hardware-in-the-Loop (HIL) lab systems, integrating Ubuntu bench machines and Raspberry Pis with Jenkins.
- Analyzed/triaged CI test failures every morning and created/updated Jira tickets to streamline development feedback loops.

UC Santa Cruz

Santa Cruz, CA

Embedded Systems Design Lab Tutor

September 2021 – November 2021

- Led 3 lab sections a week with 1-2 other tutors.
- Assisted around 20 students each section by answering questions and providing debugging advice.
- Taught students how to debug hardware with Oscilloscopes, Multimeters, LEDs, and LCD screens.
- Communicated effectively and concisely to ensure every student received assistance before the section ended.

- Helped students understand and use concepts such as RTOS, UART, DMA, ADC, DAC, PWM, etc.
- Performed check-offs and graded lab submissions while cross-referencing colleagues to ensure equal and fair grading among students.

San Jose Eyecare

Optometric Technician

San Jose, CA

June 2016 – September 2021 (Summers)

- Operated diagnostic equipment.
- Authorized and checked coverage of patient insurance.
- Scheduled appointments, took calls, handled pickups, and other front-desk tasks.

EDUCATION

University of California, Santa Cruz

*Bachelor of Science (B.S.), Computer Engineering
Systems Programming Concentration*

Santa Cruz, CA

Sep 2018 – Jun 2022

PROJECTS

Two Channel Oscilloscope | *Embedded Systems Design*

April 2021 – June 2021

- Displayed waveforms and frequencies onto an LCD screen; supported vertical and horizontal scaling.
- Read voltage values on 2 GPIO pins using an ADC; data points transferred from ADC buffer to ping-pong buffers via DMA.
- Processing done on one of the ping-pong buffers to calculate frequencies, select data points to render, and detect triggers.
- User input enabled through knobs and command line; knobs are potentiometers whose voltage is monitored by the ADC, commands are sent over UART and parsed on the microcontroller.
- Designed for the PSoC-6 microcontroller and written in C.

Quadruple Software UART | *Embedded Systems Design*

April 2021 – June 2021

- Optimization exercise to implement four UARTs in software and on just the CM4 core.
- UARTs operated concurrently as tasks in FreeRTOS.
- Written in C for the PSoC-6 micro-controller.

Pintos Modification Labs | *Principles of Computer Systems Design*

January 2021 – March 2021

- Improved and implemented aspects of the Pintos operating system, including the ability to block threads, priority-based thread scheduling, and priority donation/inheritance.
- Worked with multi-threading and concurrency primitives.
- Implemented condition variables and locks using the pre-existing semaphore implementation.
- Debugging was performed with GDB.

Prime Number Detector | *Logic Design*

October 2020 – December 2020

- Written in gate level Verilog for the Basys-3 FPGA.
- Determines whether a (16 bit) number is prime; if not, shows the least prime factor.
- Values shown on a 7-segment display.
- Implemented binary long division.

Plux: Smart Outlet | *Engineering Design Project*

January 2022 – June 2022

- Worked with a team to create an IoT device that plugs into an outlet and allows for remote switching and monitoring of power consumption.
- Programmed an ESP32 to connect to WiFi and publish/receive MQTT messages; programmed in C++ with the Arduino IDE.
- Used Python to create an MQTT client that parses messages and writes to an SQLite3 database on an Oracle Cloud instance.
- Designed the architecture and protocol for sending messages between the smart plug, server, and web-client.

Unnamed Bus System Project | *Not affiliated with UCSC*

January 2022 – June 2022

- Contributed to the creation of an integrated hardware and software system to track a transit bus's position and route, as well as provide a console for the bus driver to interact with the system.
- Designed and wrote the program flow for the driver console in C; uses a simplified state-machine design pattern that makes use of function pointers to encapsulate each state.

- Use CMake to cross compile for RP2040 on the Raspberry Pi Pico.

BAM! File Sharing Web App | *Introduction to Software Engineering*

March 2022 – June 2022

- A web app that allows devices to share files by fist-bumping; files are sent peer-to-peer via WebTorrent.
- Designed and wrote the back-end architecture and the logic for connecting two clients and sending files.
- Worked with a team using Scrum and followed Scrum principles; was also the Product Owner.
- Built with React, Express, NodeJS, and MongoDB (MERN Full-Stack).

Reliable File Transfer Protocol | *Network Programming*

January 2022 – March 2022

- Created and implemented a protocol for reliable data transfer between a server and client via UDP.
- Can handle multiple clients concurrently.
- Improved performance by using a sliding window and buffering out of order packets.

HONORS

Tau Beta Pi (TBP), Member: Santa Cruz (Fall 2022)